

GUNDLAPALLI MOULENDRA BALAJI

Hyderabad, India | +91-8074061387 | moulendrabalaji2007@gmail.com

Portfolio | LinkedIn | GitHub | LeetCode

SUMMARY

Machine Learning Engineer building applied systems in agentic AI, generative AI, and computer vision with Python, JavaScript/TypeScript, FastAPI, LangChain, and React. Authored a capstone research paper on predictive modeling for search-visibility decline using gradient-boosted trees. Brings strong DSA fundamentals from competitive programming and open-source work, with a clear focus on shipping production-ready ML products.

EDUCATION

CR Rao AIMSCS <i>B.Tech, Computer Science Engineering (AI/ML) CGPA: 8.9</i>	2024 – 2028 (Expected)
Deeksha Junior College <i>Intermediate (MPC) 93.4%</i>	2022 – 2024
Bhashyam High School <i>SSC CGPA: 9.2</i>	2016 – 2022

TECHNICAL SKILLS

AI/ML & GenAI: Large Language Models (LLMs), Generative AI, Retrieval-Augmented Generation (RAG), AI Agents and Agentic Systems, Deep Learning, Machine Learning, Vector Embeddings, Image Processing, Distributed Systems

Cloud AI & Deployment: Flask, FastAPI, Streamlit

Frameworks & Libraries: LangChain, React, Node.js, Electron.js, Three.js

Programming Languages: Python, Java, C++, JavaScript, TypeScript, HTML, CSS

Tools & Platforms: Git, GitHub, VS Code, Hugging Face, Jupyter Notebook, LaTeX, Microsoft Excel, Docker, AWS DevOps

CS Fundamentals: Data Structures & Algorithms (Design and Complexity Analysis), Object-Oriented Programming, Operating Systems, DBMS, Computer Networks

PROJECTS

GestureForge – Real-Time Gesture-Controlled 3D Editor <i>Python, OpenCV, MediaPipe, scikit-learn</i>	Sep – Dec 2025
- Built a webcam-only, gesture-controlled 3D scene editor using MediaPipe's 21-landmark hand tracking with One-Euro filtering for stable, real-time interaction. - Designed two parallel gesture-recognition engines: a scale/orientation-invariant rule-based classifier and a trainable scikit-learn RandomForest model that can be retrained live. - Implemented a software 3D renderer (model-view-projection pipeline with Lambert shading) with an optional OpenGL backend, plus physics, undo/redo, and macro recording. - Shipped with Docker and GitHub Actions CI, backed by 109 automated unit tests across camera-free modules.	
Support Triage Environment – Agentic AI Simulation <i>Python, FastAPI, Reinforcement Learning</i>	Feb – Apr 2026
- Designed an OpenEnv-style reinforcement-learning environment simulating customer-support triage, where an agent reads emails, looks up customer metadata, and searches a knowledge base to decide whether to reply or escalate. - Built a single-action command space and reward-shaped observation model across tiered difficulty levels (Easy, Medium, Hard) mirroring real support escalation policy. - Benchmarked baseline LLM-agent performance (GPT-4o) across difficulty tiers and containerized the environment with Docker behind a FastAPI-served interface.	
Conflict-Free Collaborative OLTP System <i>Python, PostgreSQL, CRDT</i>	May – Jun 2026
- Architected a conflict-free collaborative OLTP database enabling concurrent users to transact without merge conflicts, with deterministic convergence across replicas. - Implemented CRDTs and a foreign-key tombstone policy to guarantee eventual consistency, distributed uniqueness, and safe cascading deletes. - Applied CAP theorem principles to a real transactional workload, demonstrating distributed-systems theory in practice.	

RESEARCH & PUBLICATIONS

Does Query-Portfolio Diversification Predict a Page's Resilience to Visibility Decline? <i>FlyRank ML Internship Capstone</i>	Aug 2026
- Engineered a query-concentration signal, the Herfindahl-Hirschman Index, from 79M+ daily performance rows and 2.4M+ query-level records across 520K+ content pages. - Framed visibility decline as a ranking problem and applied a time-aware, page-grouped train-test split to prevent data leakage. - Trained and compared a trend-continuation baseline, logistic regression, and a LightGBM gradient-boosted model to predict which pages would lose visibility. - Achieved an AUC-ROC of 0.661 and a top-50 precision of 0.69 with the LightGBM model, a 23 percent improvement over the baseline score of 0.56. - Found query concentration to be the second most predictive feature after momentum and produced a ranked list of at-risk pages with reasons for editorial review.	

CERTIFICATIONS

DevOps Nation SkillUp – GeeksforGeeks (*Jul 2026*)

Data Visualization with Python – IBM SkillsBuild (*May 2026*)

Software Engineering Job Simulation – JPMorgan Chase, via Forage (*Jun 2026*)

Analyze Images with the Cloud Vision API – Google Cloud (*Mar 2025*)

HTML & CSS Essentials – Cisco Networking Academy (*Dec 2025*)

Machine Learning and Deep Learning Fundamentals – NPTEL (*In Progress*)

ACHIEVEMENTS & LEADERSHIP

HackerRank Orchestrate Hackathon (Global) – Ranked 158th out of 1,934 global participants (top 8%) in a 24-hour hackathon on the Orchestrate platform.

LeetCode Problem Solving – Solved 70+ DSA problems across Arrays, Dynamic Programming, Backtracking, and Hash Tables. Earned the 50 Days Consistency Badge (2026).

HackerRank Campus Crew Ambassador – Selected by HackerRank to represent the university, promote competitive-programming culture, and coordinate coding events within the student community.

Ascent Anvil Hackathon, Scaler School of Technology – Collaborated in a team under a time-bound engineering challenge, designing and shipping a working prototype end-to-end.

Open-Source Activity – Maintain 17+ public GitHub repositories (9 stars) spanning agentic AI, computer vision, distributed systems, and full-stack projects.